

SYNX - COMPLETE SYSTEM EXPLAINER & ARCHITECTURE GUIDE

Smart India Hackathon 2026 | Problem Statement ID: 26034 | Team SYNX
Automated Verification under Legal Metrology (Packaged Commodities) Rules, 2011

In Simple English: SYNX is an automated AI-powered digital inspector. It takes photos of any packaged product (like chips, cold drinks, or shampoo), reads all the text, mathematically measures the font height down to millimeters, verifies every mandatory Indian legal declaration, and instantly produces an official, court-admissible PDF violation and audit certificate.

1. THE 8 PIPELINE STAGES (DATA FLOW AT A GLANCE)

Stage	Component	Plain English Job (Analogy)
1. Ingestion	FastAPI REST API + QR Session	Takes photos from phone camera or desktop without installing apps.
2. Anti-Glare Clean	OpenCV + CLAHE Filter	Like polarized sunglasses for AI; removes shiny light glare from foil/plastic.
3. Spatial OCR	RapidOCR (PaddleOCR ONNX)	Reads every single word and records where it sits on the packaging.
4. Calibration	Millimeter Scale Calculator	Converts screen pixels into real millimeters using package dimensions or a coin.
5. Entity NLP	Heuristic Regex & Detective Brain	Figures out which word is Price, Date, Factory, or Weight (and filters nutrition).
6. 3D Assembly	Multi-Angle Aggregator	Combines Front, Back, and Cap into one single unified product brain.
7. Rulebook	Legal Metrology Rules Engine	Evaluates rules, detects violations, and calculates Sec 36(1) penalty risk.
8. Legal Report	HTML5 Canvas + ReportLab PDF	Draws green/red boxes on screen and prints an official court-ready certificate.

2. STEP-BY-STEP TECHNICAL EXPLANATIONS & JARGON BUSTERS

Step 1: Smartphone Zero-Install QR Ingestion

REST API & Polling: Think of the REST API as a waiter in a restaurant. When you scan the QR code on your phone, the server opens a private mailbox (Session ID). The phone drops photos into that mailbox, and the laptop checks the mailbox every second until the photos arrive. No native mobile app install is needed!

Step 2: Glare Attenuation & CLAHE Preprocessing

Computer Vision (OpenCV) & CLAHE: Real packaging (chips packets, plastic bottles) acts like a mirror under store lights. Ordinary cameras see blinding white glare spots. CLAHE (Contrast Limited Adaptive Histogram Equalization) cuts the image into tiny micro-tiles and balances dark/light areas locally. It works like polarized sunglasses, cutting through reflections to reveal faint dot-matrix printing.

Step 3: Spatial OCR & Bounding Box Detection

PaddleOCR ONNX & Bounding Boxes: We use a pre-trained deep neural network running on ONNX. It reads text and returns the exact pixel box (X, Y, W, H) and confidence score (e.g., 98% sure) for every word. Because it uses ONNX, it runs in less than 1 second on a standard laptop CPU without requiring an expensive gaming graphics card (GPU).

Step 4: Millimeter Calibration & Rule 7 PDP Calculations

Scale Factor & Principal Display Panel (PDP): A screen only knows pixels, not millimeters. By inputting the package dimensions or using a 5 coin (legally exactly 23.0 mm), our math engine computes the scale (e.g. 0.1 mm per pixel). It calculates the PDP area (front face of the pack) and looks up the statutory minimum font height tier under Rule 7 (e.g. >= 2.0 mm or >= 4.0 mm).

Step 5: NLP Entity Parsing & Blacklist Filtering

Regex, Blacklists & Disambiguation: OCR gives us 60 random words. Our detective brain uses regex patterns to extract entities: it finds 6-digit PIN codes, extracts dot-matrix dates (12/2026), strictly filters out nutrition tables (so Sugar: 14g is never mistaken for the pack weight!), and distinguishes total MRP (Rs. 30.00) from Unit Sale Price (Rs. 0.075/ml).

Step 6: Multi-Angle 3D Product Aggregation

Unified Packaging Ledger: Real packages are 3D! The brand is on the front, factory address is on the back, and the price is stamped on the cap. Our multi-angle aggregator tags each extraction with its source face and merges them into one complete audit record.

Step 7: Decoupled Statutory Rulebook & Penalties

JSON Rulebook & Section 36(1): Laws are stored in an editable JSON file rather than hardcoded in Python. If legal rules change, lawyers can edit the JSON in the browser with live hot-reloading. The engine automatically maps violations to Legal Metrology Act Sec 36(1) penalties (fines of ₹25,000, ₹50,000, or ₹1,00,000 / 1 year imprisonment).

Step 8: Court-Admissible PDF with SHA-256 Hashing

Cryptographic Evidence Integrity: Under Section 65B of the Indian Evidence Act, digital proof must be untampered. The system computes a SHA-256 mathematical hash of the photo. If someone edits even one pixel in Photoshop, the hash breaks. This proves digital authenticity in a court of law.

3. THE 8 MANDATORY STATUTORY RULES IN PLAIN ENGLISH

Statutory Rule	Plain English Meaning	What SYNX Verifies & Catches
Rule 6(1)(a)	Manufacturer Details	Checks complete address and verifies mandatory 6-digit postal PIN code.
Rule 6(1)(b)	Commodity Name	Identifies genuine generic name; filters out marketing slogans and process text.
Rule 6(1)(c)	Net Quantity & Units	Verifies SI metric units (g, ml, kg); penalizes illegal symbols (gms, kgs).
Rule 6(1)(d)	Mfg / Packing Date	Extracts MM/YYYY or DD/MM/YYYY dates, including faint dot-matrix cap stamps.
Rule 6(1)(e)	MRP & Taxes	Extracts total price and verifies 'inclusive of all taxes' declaration.
Rule 6(1)(e)	Unit Sale Price (USP)	Verifies per-unit rate (e.g. Rs. 0.075/ml) so consumers can compare value.
Rule 6(1)(f)	Consumer Care	Checks phone helpline and email; rejects fake 14-digit FSSAI numbers.
Rule 6(10)	Country of Origin	Checks explicit declaration (e.g. Made in India) or manufacturer postal origin.
Rule 7	Font Height Tiers	Measures physical letter height in mm against statutory minimum thresholds.
Sec 36(1)	Statutory Penalties	Calculates legal liability: ₹25k (1st), ₹50k (2nd), 1L / 1yr jail (subsequent).

How to Run in 1 Click: Double-click start.bat in the project folder to launch the server and open the browser interface automatically.